

Jordan University of Science and Technology

AI 350: Data Science

Instructor: Dr. Malak Abdullah

Project: Data Science Research Paper Project

Project Overview:

In this project, you will write a **research paper** that demonstrates your ability to apply data science techniques to solve a real-world problem. You will choose a dataset, formulate a research question, design and implement a machine learning model, and present your findings in a structured, professional research paper. This project will test your skills in data analysis, machine learning, critical thinking, and scientific writing.

Learning Objectives:

By the end of this project, you will be able to:

1. Formulate a clear and meaningful research question.
 2. Preprocess and analyze data to extract meaningful insights.
 3. Design, implement, and evaluate a machine learning model.
 4. Communicate your findings effectively in a structured research paper.
 5. Critically evaluate your work and suggest future improvements.
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Project Requirements:

1. **Dataset:**
 - Choose a publicly available dataset (e.g., from Kaggle, UCI Machine Learning Repository, or government sources).
 - The dataset should be relevant to a real-world problem and suitable for machine learning.
2. **Research Question:**
 - Formulate a clear and specific research question.
 - Example: "Can a deep learning model accurately predict customer churn based on historical data?"
3. **Methodology:**
 - Perform exploratory data analysis (EDA) to understand the dataset.
 - Preprocess the data (e.g., handle missing values, encode categorical variables, normalize data).
 - Select and implement at least three machine learning algorithms (e.g., regression, classification, clustering).

- Evaluate the model using appropriate metrics (e.g., accuracy, precision, recall, F1-score, RMSE).
 - 4. **Research Paper:**
 - Write a **7-10 page research paper** (excluding references) in the following structure:
 1. **Abstract:** Summarize the problem, methodology, key results, and conclusion.
 2. **Introduction:** Provide background, state the research question, and explain the significance of the study.
 3. **Literature Review:**
 1. Summarize key research papers related to your topic.
 2. Identify gaps and justify your research question.
 3. Use at least **10** peer-reviewed sources.
 4. **Methodology:**
 1. Clearly define the research question and hypothesis.
 2. Describe data sources, preprocessing steps, and model selection.
 3. Justify algorithm choices and evaluation metrics.
 5. **Experiments & Analysis:**
 1. Implement a small-scale experiment or case study.
 2. Present results using appropriate statistical and visual tools.
 3. Present and interpret the results using tables, graphs, and metrics.
 4. Compare findings with previous research.
 6. **Discussion & Conclusion:**
 1. Interpret results, highlighting limitations and implications.
 2. Suggest areas for future research.
 3. Discuss the implications of your findings, compare them to existing work, and address limitations.
 4. Summarize key findings and suggest future work.
 7. **References:** Cite all sources using a consistent citation style (e.g., APA, IEEE).
 - 5. **Code and Documentation:**
 - Submit a well-documented Jupyter Notebook or Python script that includes:
 - Data preprocessing steps.
 - Model implementation and evaluation.
 - Comments explaining each step of the process.
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Submission Guidelines:

1. **Research Paper:** Submit a PDF document (7-10 pages, excluding references).
 2. **Code:** Submit a Jupyter Notebook or Python script with comments.
 3. **Dataset:** Include a link to the dataset or submit it as a separate file.
 4. **Video**
 5. **Deadline: 24-4-2026**
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Example Topics:

1. Predicting house prices using regression models.
 2. Classifying spam emails using natural language processing (NLP).
 3. Clustering customers for market segmentation.
 4. Predicting student performance using educational data.
 5. Analyzing sentiment in product reviews.
 6. Bias and Fairness in Machine Learning
 7. Ethical Considerations in AI
 8. Time Series Analysis for Financial Forecasting
 9. Advances in Natural Language Processing
 10. Computer Vision in Medical Diagnostics
 11. Explainable AI and Model Interpretability
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Tips for Success:

1. Start early to allow time for data exploration, model tuning, and writing.
 2. Use tools like Grammarly for proofreading and Zotero for managing citations.
 3. Seek feedback from peers or instructors during the drafting process.
 4. Focus on clarity and reproducibility in your code and writing.
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Below are the detailed instructions and deadlines for each component of the project:

1. Research Proposal (5 points)

Deadline: March 17

Your research proposal should outline the key aspects of your project. It must include the following 5 points:

1. **Research Title:** A clear and concise title reflecting your research topic.
2. **Research Question/Objective:** Define the problem you aim to solve or the question you seek to answer.
3. **Dataset:** Describe the dataset(s) you plan to use, including its source and relevance to your research.
4. **Methodology:** Briefly explain the techniques, algorithms, or tools you intend to use for analysis.
5. **Expected Outcome:** State the potential results or contributions of your research.

Submit your proposal as a **1-2 page document** in PDF format.

2. Introduction and Literature Review

Deadline: April 10

This section of your research paper should include:

1. **Introduction:** Provide background information on your topic, its significance, and the motivation behind your research.
2. **Literature Review:** Summarize and analyze existing research related to your topic. Highlight gaps in the literature that your research aims to address.

Submit this section as a **3-5 page document** in PDF format, including proper citations and references.

3. Full Research Paper + Code (15 points)

Deadline: April 24

Your final research paper should include the following sections:

1. **Introduction** (revised based on feedback)
2. **Literature Review** (revised based on feedback)
3. **Methodology:** Detailed explanation of your approach, including data preprocessing, analysis, and tools used.
4. **Results:** Present and interpret your findings using tables, graphs, or visualizations.
5. **Discussion:** Analyze the implications of your results and compare them with existing studies.
6. **Conclusion:** Summarize your research, its limitations, and potential future work.
7. **References:** Include all sources cited in your paper.

Submit the full paper as a **7-10 page document** in PDF format.

4. Presentation or video (5 points)

Deadline: May 1st

Along with your research paper, you are required to submit:

1. **Code:** Submit all scripts, notebooks, or programs used for your analysis. Ensure your code is well-documented and organized.
2. **Presentation:** Create a **10-minute presentation** summarizing your research. Include slides covering the problem statement, methodology, results, and conclusions.

Submit your code as a **ZIP file** and your presentation as a **PDF or PowerPoint file**.

Grading Breakdown

- Research Proposal: 5 points
 - Introduction and Literature Review: 0 points
 - Full Research Paper + code : 15 points
 - Video Presentation: 5 points
- Total: 25 points**
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Important Notes

- Late submissions will incur a penalty of **2 points per day**.
- Ensure all submissions are original and properly cited to avoid plagiarism.
- Use clear and professional language in your writing.

If you have any questions or need clarification, feel free to reach out to me. I look forward to seeing your innovative research ideas and results!